Tabular Foundation Models in the Low-Data Regime: A Two-Stage Benchmark of Predictive Performance and Computational Cost

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Abstract—*Tabular foundation models (TFMs) use knowledge acquired through pretraining to support prediction from a small labelled context, but the extent to which this advantage persists as task-specific data increase, and whether it justifies the associated computational cost, remains an empirical question. We present a two-stage benchmark on OpenML-CC18 binary-classification tasks. Stage I compares two complementary TFM designs—table-native TabPFN-v2 and semantics-aware SAP-RPT-1, which incorporates semantic information from column headers and text-valued cells—with XGBoost, LightGBM, CatBoost, and a multilayer perceptron across 15 datasets. Based on paired ROC-AUC results, TabPFN-v2 is selected for the detailed scaling analysis. Stage II evaluates it against four conventional baselines across eight datasets, nine training sizes (N = 25 to the full training split), and five random seeds, yielding 1,800 model-dataset-seed-size evaluations. Conventional baselines undergo hyperparameter tuning with Optuna, whereas TabPFN-v2 is evaluated zero-shot. TabPFN-v2 records the highest mean ROC-AUC at all nine sizes. Its advantage over the best baseline is largest at N = 25 (0.0346 AUC) and narrows overall to 0.0087 at the full training size. Repeated paired tests support the observed differences, while a label-permutation diagnostic reduces performance to approximately chance, supporting dependence on the feature-label relationship. Dataset-level crossover points nevertheless vary widely. The broader evaluation also shows that higher discrimination does not guarantee the best PR-AUC, calibration, memory use, or inference efficiency. These findings support evaluating TFMs across data regimes, datasets, predictive metrics, and deployment costs rather than relying on a single aggregate score.*

Keywords—tabular foundation models; low-data learning; few-shot learning; in-context learning; OpenML-CC18; ROC-AUC; model efficiency.

1. INTRODUCTION

Tabular data are a core input to predictive systems in finance, healthcare, business analytics, public services, and scientific research. Unlike images or natural-language sequences, tabular datasets combine columns with different meanings, scales, distributions, and data types. They are often assembled for a specific task and may contain relatively few labelled examples, class imbalance, missing observations, and domain-dependent feature relationships. Reliable learning from limited labels is therefore not a peripheral use case: it is a recurring practical requirement for tabular machine learning.

Gradient-boosted decision trees (GBDTs), particularly XGBoost, LightGBM, and CatBoost, remain strong general-purpose approaches for supervised tabular prediction [6]-[8]. Multilayer perceptrons (MLPs) provide a neural alternative. These conventional models are typically fitted separately for each task, so their predictive performance depends directly on the labelled observations available for that task and on the quality of task-specific training and tuning. They remain difficult baselines to improve upon, which makes claims about a new class of tabular methods especially important to test against well-tuned implementations rather than default configurations.

Tabular foundation models (TFMs) introduce a different learning strategy. Instead of fitting all predictive structure from scratch for each new dataset, they acquire priors or representations across collections of tabular tasks during pretraining and use a task-specific labelled context when producing predictions. The TabPFN family is a prominent example: it applies a prior-data fitted network to tabular prediction and uses in-context examples rather than conventional task-specific parameter fitting [1], [2]. Recent advances in TabPFN-v2 and the emergence of other TFM approaches, together with broader comparative efforts such as TabArena, have increased interest in whether pretrained tabular priors can provide practical benefits across a range of real-world datasets [1], [4].

The potential benefit is particularly relevant when labels are scarce. A model that has acquired useful structure during pretraining may produce strong predictions from a small labelled context, reducing the amount of task-specific supervision needed to reach a given level of discrimination. However, a result measured at only one training-set size cannot reveal whether the advantage is concentrated in the few-shot regime, persists as more labels become available, or diminishes as conventional learners acquire sufficient task-specific data. Likewise, a benchmark that reports only ROC-AUC may overlook operational constraints that determine whether a model can be deployed.

These issues motivate a systematic scaling study. We ask not simply which model has the largest average score, but how relative model performance changes as labelled data increase, how consistent that pattern is across datasets, and what computational and calibration costs accompany the predictive gains. A useful assessment must consider predictive discrimination together with precision-recall behaviour, probability calibration, memory consumption, and inference time. This broader view matters because two models with similar ROC-AUC can have very different deployment profiles, while a model with the highest ROC-AUC may be unsuitable where latency or memory budgets are strict.

We address these questions through two sequential stages. Stage I establishes an empirical basis for selecting the TFM to study in depth: TabPFN-v2 and SAP-RPT-1 are compared with three GBDT methods and an MLP across 15 OpenML-CC18 binary-classification datasets. Stage II then holds the selected TFM fixed and evaluates it against tuned conventional baselines across eight datasets, nine training-set sizes, and five random seeds. The design yields 1,800 paired model-dataset-seed-size evaluations and allows the performance gap to be followed across a controlled range of label availability.

The study makes four contributions. First, it provides a two-stage comparison that makes the choice of the representative TFM empirical rather than assumed. Second, it characterizes low-data scaling over nine training sizes with repeated, paired subsamples across multiple datasets. Third, it examines whether the point at which a conventional learner matches or exceeds the selected TFM is consistent across datasets. Fourth, it evaluates predictive performance alongside calibration, memory, and inference cost, providing a more deployment-aware interpretation than a single-metric ranking. The results are specific to the evaluated binary-classification tasks and do not imply universal superiority for TFMs or a universal sample-size threshold.

2. RELATED WORK

Tabular foundation models extend the foundation-model paradigm to structured prediction tasks by learning from collections of tabular problems and using task examples as context at inference time. The original TabPFN work framed this approach through prior-data fitted networks and Bayesian-style inference [2]. TabPFN-v2 expanded the capabilities of the approach and reported strong predictive performance on small tabular datasets [1]. These developments have encouraged evaluation beyond isolated demonstrations, particularly against well-established tree-based approaches.

SAP-RPT-1 provides another representative TFM for comparative evaluation, while TabArena offers a broader benchmark of tabular models and datasets [4]. The present study complements broad model-ranking work by varying the number of task-specific labelled observations systematically. Its focus is the shape of the performance curve as N increases, not only the score obtained from a fixed training split. The analysis also relates this curve to memory, inference time, and probability calibration so that the predictive results can be interpreted in the context of practical model selection.

OpenML-CC18 provides a curated collection for reproducible classification comparisons [5]. The conventional methods used here represent strong and widely adopted tabular baselines: XGBoost [6], LightGBM [7], and CatBoost [8], alongside an MLP. Optuna is used for model-specific hyperparameter tuning of the conventional learners [9]. Finally, a label-permutation diagnostic is included as a sanity check. Permuting labels disrupts the feature-label relationship and helps assess whether predictive performance depends on the supervised signal; it does not, by itself, establish a causal explanation of a model's internal computations [10].

3. STAGE I: BROAD BENCHMARK AND TFM SELECTION

3.1 Objective and comparison design

Stage I addresses the model-selection question that precedes the main scaling analysis: which of the evaluated TFMs provides the strongest empirical basis for a detailed low-data comparison? Rather than selecting TabPFN-v2 solely because it is a well-known model, we compare it directly with SAP-RPT-1 and with conventional learners on a wider set of tasks. The benchmark includes 15 OpenML-CC18 binary-classification datasets and six models: TabPFN-v2, SAP-RPT-1, XGBoost, LightGBM, CatBoost, and an MLP.

The two TFMs were chosen to represent complementary approaches to tabular context. TabPFN-v2 follows a table-native, value- and structure-oriented formulation: it processes tabular cell values and their column structure, but does not explicitly use the natural-language meaning of feature names as semantic input. SAP-RPT-1, introduced in the ConTextTab work, is semantics-aware: its representation incorporates column headers and modality-specific semantic representations of text-valued cells [3]. Including both models therefore brings a table-native approach and a schema- and cell-semantics-aware approach into the same broad comparison. This is a comparison of complementary TFM designs, not a controlled ablation of column-name semantics alone; the models also differ in architecture, pretraining data, and other implementation choices.

The first stage is a screening and selection exercise rather than the low-data scaling experiment itself. All six candidates are assessed on the same task suite, with a common held-out test partition within each dataset and five seed-level runs. The two TFMs are evaluated zero-shot, while the conventional learners are tuned before held-out scoring. ROC-AUC provides the primary basis for comparing ranking discrimination across tasks; PR-AUC is retained as a complementary measure because the positive-class proportions differ between datasets. After this broad comparison, TabPFN-v2 and SAP-RPT-1 are also compared directly using paired dataset-seed observations. The stronger paired ROC-AUC result determines which TFM is carried into Stage II. This step selects a model for deeper study; it does not establish dominance across every metric or deployment requirement.

For a given dataset, the models are assessed on a common held-out test partition so that the resulting scores are comparable within the dataset. Five seed-level runs are used to assess variation across runs. The two TFMs are evaluated zero-shot, without task-specific hyperparameter search, whereas the conventional baselines undergo model-specific hyperparameter tuning using Optuna [9]. This arrangement compares pretrained inference with a practical, tuned conventional-learning workflow; it is not intended to represent exhaustive optimization of every baseline.

Stage I considers ROC-AUC as the primary discrimination measure and also records PR-AUC and selected operational measures, including peak memory and inference time per test row. Including PR-AUC is important because the datasets differ in positive-class prevalence, and ROC-AUC alone does not fully describe precision-recall behaviour. Resource measures provide an initial view of whether the predictive ranking is accompanied by meaningful differences in deployment cost.

3.2 Comparative results and selection

TabPFN-v2 obtains the highest mean ROC-AUC across the 15 datasets (0.9101) and ranks first on 10 of them. SAP-RPT-1 records a mean ROC-AUC of 0.8906, while the three GBDT baselines range from 0.8751 to 0.8787 and the MLP records 0.8532. The PR-AUC ranking is not identical: SAP-RPT-1 records 0.7501, compared with 0.7269 for TabPFN-v2. This distinction is important. TabPFN-v2 is strongest on average ROC-AUC, but the Stage-I evidence does not show that it leads on every predictive metric. Table I summarizes these predictive and operational results.

TABLE I. STAGE-I PERFORMANCE AND RESOURCE SUMMARY

Model	ROC-AUC	PR-AUC	Peak memory (MB)	Inference (ms/row)
TabPFN-v2	0.9101	0.7269	1,494.76	65.1142
SAP-RPT-1	0.8906	0.7501	2,532.02	4.180
XGBoost	0.8765	0.7065	590.88	0.0397
LightGBM	0.8751	0.6966	536.88	0.0246
CatBoost	0.8787	0.7086	637.88	0.0105
MLP	0.8532	0.6723	571.70	0.0120

The direct TabPFN-v2 versus SAP-RPT-1 comparison contains 75 paired dataset-seed observations. TabPFN-v2 records the higher ROC-AUC in 57 of the 75 pairs, with a mean paired difference of +0.0214. The paired bootstrap 95% confidence interval is [0.0111, 0.0342]. The Wilcoxon signed-rank p-value is 4.96 × 10^-10 and the Holm-adjusted value is 1.98 × 10^-9; Cliff's delta is +0.587. Together, these results provide a statistically supported basis for carrying TabPFN-v2 into Stage II. The selection is empirical and tied to ROC-AUC performance under this benchmark; it is not a claim that TabPFN-v2 is the best TFM for every metric, dataset, or deployment setting.

The operational columns in Table I make the selection trade-off more concrete. TabPFN-v2 records 65.1142 ms per test row, compared with 4.180 ms for SAP-RPT-1 and 0.0105-0.0397 ms for the GBDT methods. Its peak memory is 1,494.76 MB: lower than SAP-RPT-1 but higher than the three GBDT baselines. Thus, Stage I does not identify a model that dominates every dimension. TabPFN-v2 is carried forward because of its stronger paired ROC-AUC performance, while the measured latency and memory costs remain part of the interpretation of the Stage-II results.

4. STAGE II: LOW-DATA SCALING EVALUATION

4.1 Objective and dataset selection

Stage II investigates the central research question: how does the predictive performance and computational behaviour of the selected TFM change as the amount of labelled task-specific data increases, and how does that behaviour compare with tuned conventional learners? To answer this question, the evaluation varies training-set size while keeping the held-out test data fixed within each dataset. This isolates changes associated with the number of labelled training observations from changes in the test partition.

Datasets were selected from OpenML-CC18 [5] using four criteria. Tasks had to be binary-classification problems; contain more than 1,000 training observations after an 80/20 stratified split; retain at least 100 held-out test observations; and contribute no more than one dataset from any application domain. These criteria yield eight datasets spanning finance, security, software engineering, environmental monitoring, audio, healthcare, text-derived features, and historical data. Table II reports the dataset identifiers, domain labels, post-split sample counts, feature counts, and positive-class rates measured on the full training portion.

TABLE II. STAGE-II DATASETS AND CHARACTERISTICS

Dataset (OpenML ID)	Domain	Train / test rows	Features	Positive rate
Australian (40536)	Finance	6,702 / 1,676	120	0.165
Banknote Auth. (1462)	Security	1,097 / 275	4	0.445
KC1 (1067)	Software	1,687 / 422	21	0.155
Ozone Level (1487)	Environment	2,027 / 507	72	0.063
Phoneme (1489)	Audio	4,323 / 1,081	5	0.294
Sick (1050)	Healthcare	3,017 / 755	29	0.102
Spambase (44)	Text features	3,680 / 921	57	0.394
Titanic (40945)	Historical	1,047 / 262	13	0.382

4.2 Training-size grid and paired sampling

The Stage-II training-size grid contains nine conditions: N ∈ {25, 50, 100, 200, 300, 500, 750, 1,000, full}. The smallest condition represents a strict low-data setting; the intermediate conditions show how predictive performance changes as labelled examples accumulate; and full uses the complete training portion available after the train-test split. The largest possible N therefore differs by dataset, as reflected by the sample counts in Table II.

For each dataset, an 80/20 stratified train-test split is created using random_state = 42. The held-out test set is fixed across all models, training sizes, and seeds. Five seeds ({0, 1, 2, 3, 4}) are used to create training subsamples without replacement while preserving class representation. Crucially, every model receives the same sampled training rows within a given dataset-seed-size condition. Model comparisons are therefore paired on both the test observations and the labelled examples supplied for fitting or in-context prediction.

The full condition requires a careful interpretation because it represents each dataset's complete post-split training portion, not one common absolute sample size across all eight tasks. For example, the full training set ranges from 1,047 observations for Titanic to 6,702 for Australian. The fixed-N comparisons therefore provide the clearest view of behaviour at matched label counts, whereas the full condition summarizes each dataset at its maximum available training size.

The five seeds vary the sampled training subset rather than the test partition. This repeated-subset design helps quantify sensitivity of the model comparison to which labelled examples are available at a given N. It does not measure variation arising from alternative held-out test splits, a distinction that is relevant when interpreting the confidence intervals and statistical tests.

4.3 Models and tuning protocol

Five models are included in each Stage-II condition: TabPFN-v2, XGBoost, LightGBM, CatBoost, and an MLP. TabPFN-v2 is evaluated zero-shot, with n_estimators = 8 and max_context_size = 8,000, and receives no task-specific hyperparameter search. The conventional baselines (XGBoost, LightGBM, CatBoost, and MLP) undergo model-specific hyperparameter tuning using Optuna [10]. Three-fold stratified cross-validation is used where the available minority-class count permits it. For Sick and Ozone Level at N = 25, a single-fit fallback is used because the class counts do not support the planned stratified folds. This protocol compares pretrained inference with conventional learners that are tuned for the available training context.

The model-specific Optuna configurations are held consistent across the experiment to make the baseline tuning procedure explicit and repeatable. The searches are not exhaustive, and

the results should be interpreted as a comparison against tuned conventional learners under the stated protocol. The comparison is intentionally asymmetric: TabPFN-v2 uses its pretrained model without task-specific tuning, while conventional learners receive dataset- and condition-specific search. This reflects the practical contrast between pretrained inference and task-specific fitting, while making the tuning procedure an important part of the study design.

4.4 Metrics and statistical analysis

ROC-AUC on the fixed held-out test set is the primary metric because it summarizes ranking discrimination across classification thresholds. The benchmark also records PR-AUC, Brier score, log-loss, balanced accuracy, expected calibration error with 10 bins (ECE-10), peak memory, and inference time per test row. PR-AUC complements ROC-AUC by emphasizing precision and recall, which is particularly informative when positive-class prevalence is low. Brier score, log-loss, and ECE-10 assess the quality of predicted probabilities rather than ranking alone. Peak memory and per-row inference time characterize resource demands that can determine whether a predictive gain is operationally usable. Raw probabilities are evaluated without post-hoc calibration. These measures are reported separately because they describe distinct properties; without an explicit application-specific utility function, combining them into a single score would obscure rather than resolve the trade-offs.

TabPFN-v2 is compared with each conventional baseline using paired Wilcoxon signed-rank tests. Holm-Bonferroni correction is applied across the four baseline comparisons within each declared analysis scope: separately at each training size and within each dataset. Cliff's delta is reported as an effect-size measure, and paired percentile bootstrap confidence intervals use 2,000 resamples. A Friedman omnibus test with Nemenyi post-hoc comparisons provides an additional non-parametric comparison across the five Stage-II models.

The pairing scope differs by analysis. Each per-size comparison contains 40 paired observations (eight datasets × five seeds), while each per-dataset comparison contains 45 observations (nine training sizes × five seeds). These paired observations are not independent replicates of a population of datasets: the same eight datasets and held-out partitions recur across training sizes. Statistical evidence is therefore interpreted within the defined benchmark and sampling design, rather than as proof of universal performance across all tabular problems.

A label-permutation diagnostic is included as a negative control because the strong low-data ROC-AUC results could otherwise be questioned as a chance pattern, a peculiarity of a sampled subset, or an artefact of the evaluation setup. The diagnostic shuffles the label assignments while retaining the feature observations, thereby breaking the original feature-label correspondence. If the model continued to achieve substantially above-chance discrimination under this condition, the low-data result would require further scrutiny because the apparent performance would not depend on the intended supervised relationship. Instead, performance falls to approximately chance after permutation. This supports the narrower interpretation that the unshuffled results depend on a meaningful relationship between the input features and target labels. The diagnostic is considered together with the repeated-seed and paired statistical analyses; it does not by itself rule out every possible evaluation artefact, establish causality, or identify the model’s internal learning mechanism.

5. STAGE-II RESULTS

5.1 Predictive performance across training sizes

TabPFN-v2 records the highest mean ROC-AUC at every one of the nine evaluated training sizes. Table III reports the mean score for TabPFN-v2, the strongest conventional baseline at each size, their AUC difference, the Holm-adjusted p-value, and Cliff's delta. Figure 1 plots the mean ROC-AUC curves across the eight datasets and five seeds. At N = 25, TabPFN-v2 achieves a mean ROC-AUC of 0.8469, compared with 0.8123 for the strongest baseline, giving a difference of 0.0346. The gap is smaller at N = 50 (0.0230) and N = 100 (0.0129), and remains narrower than the low-data gap at the largest settings: 0.0072 at N = 1,000 and 0.0087 at full training size.

The advantage declines overall as training data increase, although the decrease is not strictly monotonic at every adjacent setting: for example, the difference is 0.0129 at N = 100 and 0.0134 at N = 200. The broad pattern is nevertheless clear. TabPFN-v2's largest aggregate advantage appears when the labelled context is smallest, and conventional learners move closer as they receive more task-specific examples. Importantly, the advantage does not disappear in the aggregate at full training size; the mean difference remains 0.0087.

All nine per-size comparisons against the strongest baseline are statistically significant after the reported Holm correction. Cliff's delta is positive at every size, ranging from +0.40 to +0.78 in Table III. Thus, the aggregate low-data result is not based on a single dataset or one favourable random subsample. The paired design, five seeds, and multiplicity-adjusted tests provide complementary evidence for the observed difference within the selected benchmark. Figure 1 also shows that the relative distance between model curves becomes smaller at larger N, while dataset-level variation remains visible in the shaded bands.

TABLE III. STAGE-II ROC-AUC BY TRAINING SIZE

N	TabPFN-v2 AUC	Best baseline AUC	Delta AUC	Holm-adjusted p	Cliff's delta
25	0.8469	0.8123	0.0346	2.14 × 10^-3	+0.40
50	0.8610	0.8380	0.0230	3.55 × 10^-4	+0.65
100	0.8915	0.8785	0.0129	5.83 × 10^-3	+0.55
200	0.9160	0.9026	0.0134	2.23 × 10^-5	+0.70
300	0.9184	0.9083	0.0100	7.91 × 10^-5	+0.72
500	0.9291	0.9200	0.0091	1.85 × 10^-5	+0.70
750	0.9353	0.9269	0.0084	7.91 × 10^-5	+0.65
1,000	0.9392	0.9320	0.0072	5.03 × 10^-5	+0.78

N	TabPFN-v2 AUC	Best baseline AUC	Delta AUC	Holm-adjusted p	Cliff's delta
Full	0.9539	0.9452	0.0087	2.14 × 10^-3	+0.57

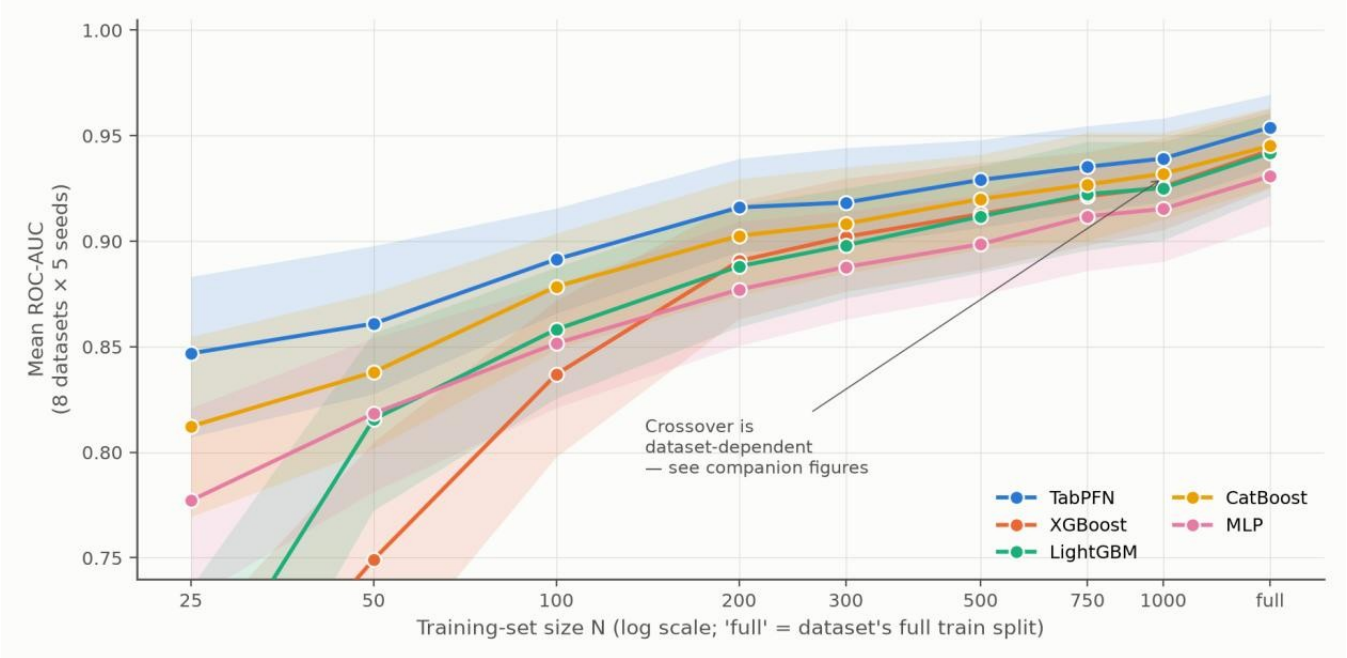


Fig. 1. Mean ROC-AUC by training-set size, averaged over eight datasets and five seeds. Shaded bands show variability across the evaluated dataset-seed conditions.

5.2 Dataset-level crossover heterogeneity

An aggregate mean curve can conceal substantial differences between datasets. To characterize this variation, we define the crossover size N* as the first evaluated training size at which the best-performing GBDT reaches or exceeds TabPFN-v2's mean ROC-AUC for that dataset. Table IV reports the best GBDT at full training size, the difference between TabPFN-v2 and the best GBDT at full N, and estimated crossover size with its bootstrap interval. The full-N difference is defined as TabPFN-v2 AUC minus the best GBDT AUC; negative values indicate that the GBDT is higher.

The observed crossover points differ markedly. The best GBDT matches or exceeds TabPFN-v2 at N = 25 for Phoneme and Sick, at N = 50 for Australian, and at N = 750 for Banknote Authentication. No crossover is observed within the available training-size range for KC1, Ozone Level, Spambase, or Titanic. For these four datasets, the reported N* is right-censored at the full training size: it indicates that no crossover was observed in the measured range, not that a future crossover is predicted.

This heterogeneity qualifies the aggregate result. Although TabPFN-v2 leads in mean ROC-AUC at every training size when results are averaged across all eight tasks, the best conventional learner can catch up quickly on some datasets and not within the measured range on others. The bootstrap intervals are broad for several tasks, so the crossover estimates should be read as descriptive summaries of evaluated conditions rather than precise universal thresholds. A model-selection rule based only on N would therefore discard material information about dataset characteristics.

TABLE IV. DATASET-LEVEL CROSSOVER ANALYSIS

Dataset	Best GBDT (AUC)	Delta AUC at full N	Crossover N* (95% CI)
Australian	XGBoost (0.8881)	-0.0197	50 (50, —)
Banknote Auth.	LightGBM (1.0000)	0.0000	750 (300, 750)
KC1	CatBoost (0.8271)	+0.0386	>= 1,687 (full)
Ozone Level	CatBoost (0.9306)	+0.0150	>= 2,027 (full)
Phoneme	CatBoost (0.9516)	+0.0165	25 (25, 500)
Sick	CatBoost (0.9998)	0.0000	25 (25, 100)
Spambase	LightGBM (0.9872)	+0.0039	>= 3,680 (full)
Titanic	LightGBM (0.9868)	+0.0059	>= 1,047 (full)

5.3 Computational cost and calibration

Predictive performance is only one part of the comparison. Figure 2 reports median peak memory and median inference time per row as functions of training-set size. The plotted TabPFN-v2 memory requirement is consistently above those of the conventional baselines and remains around the gigabyte scale across the evaluated settings; at full training size, the figure annotates a memory level of approximately 13 times the smallest baseline. The baseline memory curves are comparatively stable. This makes memory capacity a relevant

constraint when considering the largest available context, even when the predictive score is favourable.

The inference-time curve shows a more pronounced separation. TabPFN-v2 inference time rises with context size, increasing by approximately seven-fold between N = 25 and the full training condition as annotated in Fig. 2(b). In contrast, the GBDT and MLP inference curves remain close to the low-latency region and comparatively stable across N. The absolute timings are implementation- and hardware-dependent, but the relative pattern is operationally important: the model with the highest mean ROC-AUC is not necessarily the appropriate choice for a latency-sensitive prediction service.

These measurements matter because the context size is not only an experimental axis: it is also a computational input to the prediction pipeline. A memory-intensive model may restrict the context that can be processed on a given machine, while higher per-row inference time can become decisive when predictions are served at scale. Conversely, a latency difference may be less important in an offline batch-scoring workflow where discrimination is the primary objective. Reporting memory and inference alongside predictive metrics therefore makes the benchmark more informative for deployment decisions than a leaderboard based on ROC-AUC alone.

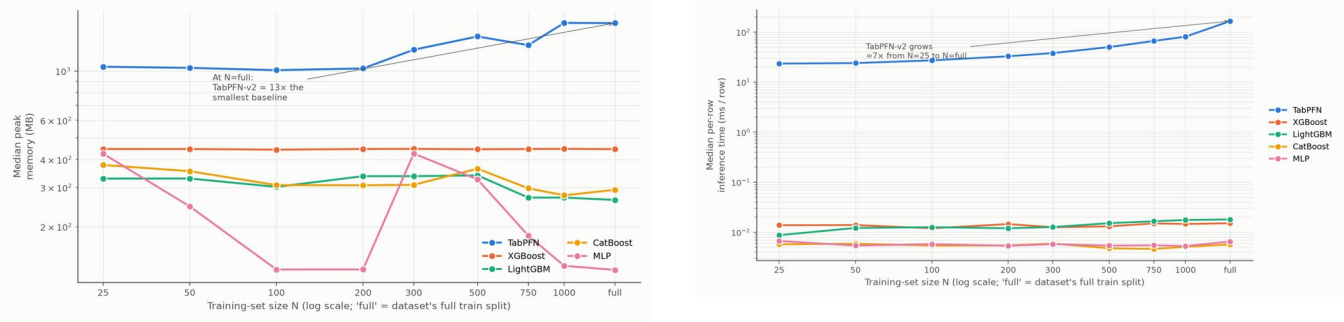


Fig. 2. Resource scaling with training context: (a) median peak memory; (b) median per-row inference time.

Figure 3 places mean ROC-AUC against inference time for each model-training-size condition. The resulting Pareto view makes the performance-efficiency tension explicit: conventional learners occupy the low-inference-cost region, whereas TabPFN-v2 provides higher aggregate ROC-AUC at substantially greater inference cost. The Pareto frontier therefore represents different operating points rather than a single universally preferred model. A choice on that frontier depends on the relative value assigned to discrimination and response time.

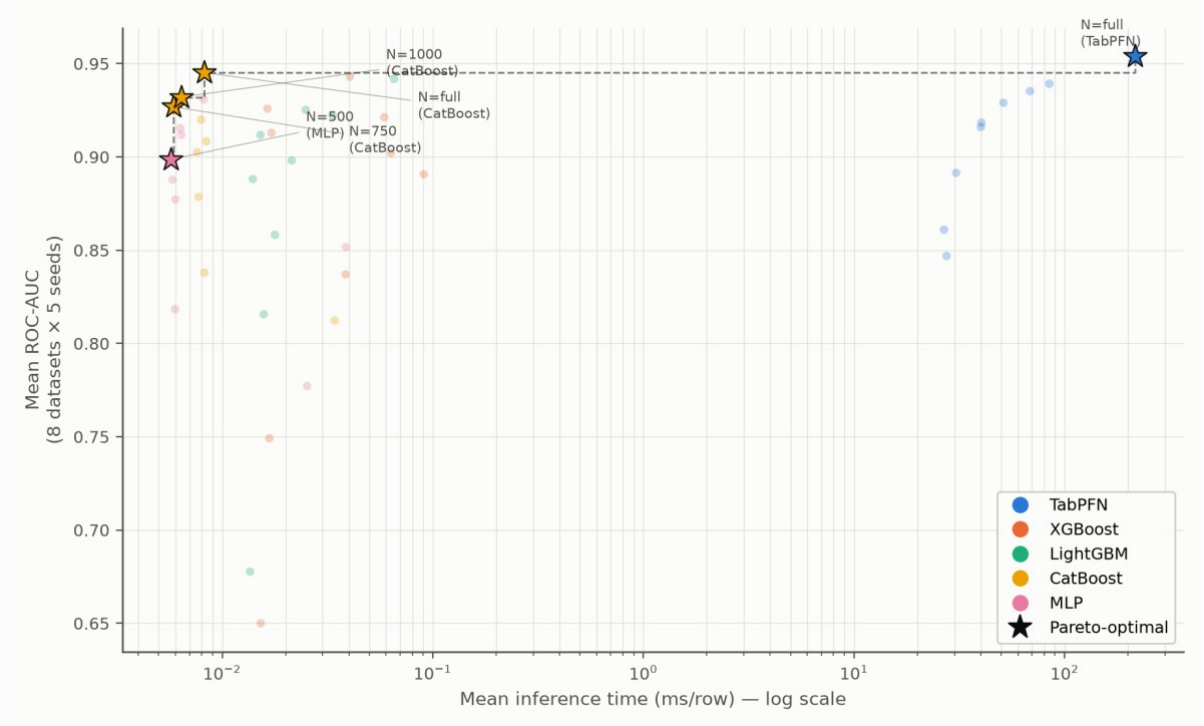


Fig. 3. Mean ROC-AUC versus inference time per test row for each model-training-size condition. The Pareto frontier identifies non-dominated aggregate operating points.

Calibration provides a further qualification. Table V reports mean ECE-10 at four representative training sizes. ECE-10 decreases with increasing N for all five models. TabPFN-v2 improves from 0.180 at N = 25 to 0.060 at full training size, but it is not the best-calibrated model among the values shown. CatBoost has the lowest ECE-10 at each reported size, falling from 0.140 to 0.040. These values are computed from raw probabilities without post-hoc calibration, so they compare the models' unadjusted probability outputs. The result illustrates why discrimination and calibration should be considered separately: the model that ranks observations best need not produce the best-calibrated probabilities.

TABLE V. MEAN ECE-10 BY TRAINING SIZE

N	TabPFN-v2	XGBoost	LightGBM	CatBoost	MLP
25	0.180	0.160	0.180	0.140	0.200
100	0.130	0.100	0.120	0.090	0.160
500	0.090	0.070	0.070	0.060	0.120
Full	0.060	0.050	0.050	0.040	0.100

The secondary measures serve different interpretive purposes and should not be collapsed into a single composite score without an explicit decision rule. PR-AUC places emphasis on precision and recall, which can be especially informative on tasks with low positive-class prevalence; the Stage-I values already show that its ranking can differ from ROC-AUC. Brier score and log-loss assess probabilistic accuracy, while ECE-10 summarizes the difference between predicted confidence and empirical outcome frequency across confidence bins. Memory and inference time describe resource requirements rather than predictive quality. In Stage II, the main-text numerical summaries focus on ROC-AUC, ECE-10, memory, and inference; the other recorded per-cell metrics should accompany the benchmark release so that readers can inspect them without treating an unreported value as evidence for a particular model ranking.

5.4 Label-permutation diagnostic

Consistent with the negative-control rationale, the label-permuted condition reduces TabPFN-v2 performance to approximately chance level. The result indicates that the model’s high discrimination in the original experiments is not reproduced when the feature-label association is disrupted. It therefore provides a diagnostic check that the low-data advantage is tied to supervised signal in the evaluated data, rather than being a score that persists independently of the target relationship. This finding strengthens the interpretation of the benchmark when read alongside the five-seed paired comparisons and multiplicity-adjusted tests, while remaining distinct from a causal explanation of the model’s internal computation.

6. DISCUSSION

6.1 Interpreting the low-data advantage

The central finding is that TabPFN-v2's aggregate ROC-AUC advantage is most pronounced when labelled data are scarce. The difference between TabPFN-v2 and the strongest conventional baseline is 0.0346 at N = 25 and 0.0087 at the full training size. The narrowing gap is consistent with the practical motivation for pretrained tabular models: when only a small labelled context is available, a model pretrained across other tasks can provide a useful starting point. As more task-specific labels become available, conventional learners can fit the target dataset more effectively and move closer in predictive performance. The current experiments support this empirical interpretation, but they do not isolate pretraining as the sole cause of the observed advantage.

The fact that the gap contracts overall should not be mistaken for a fixed switching rule. Dataset-level analysis shows that the crossover can occur at the smallest evaluated size, at an intermediate size, or not within the available training range. These differences suggest that the value of a pretrained prior depends not only on N but also on the structure and difficulty of the particular dataset. For this reason, a practitioner should validate the candidate methods on the target data and should not infer a universal threshold from the aggregate curve.

The scale of the aggregate gap should also be interpreted alongside the absolute scores. At N = 25, the difference of 0.0346 is the largest observed separation and is the condition in which the value of a pretrained model is most visible. By N = 1,000, the difference has narrowed to 0.0072 while both TabPFN-v2 and the strongest baseline exceed 0.93 mean ROC-AUC. At full training size, both exceed 0.94 and the difference is 0.0087. The practical value of these remaining differences depends on the cost of a ranking error in the application. Statistical significance indicates that the paired differences are systematic within this benchmark; it does not, on its own, establish that every application would obtain a material operational benefit from the same AUC margin.

6.2 Why a multi-objective evaluation matters

The broader metric set changes the interpretation of the predictive ranking. Stage I shows that TabPFN-v2 has the highest mean ROC-AUC, but SAP-RPT-1 has higher mean PR-AUC in the reported comparison. In Stage II, TabPFN-v2 remains competitive in calibration but does not have the lowest ECE-10; CatBoost is better on the reported ECE values. The resource plots add another distinction: TabPFN-v2's inference time grows with context size and remains markedly higher than that of the conventional methods. These observations do not weaken the AUC result. Rather, they establish that predictive discrimination, precision-recall behaviour, probability calibration, memory, and latency answer different questions and should be considered together when selecting a model.

The numerical examples underline this point. In Stage I, the difference in mean PR-AUC between SAP-RPT-1 (0.7501) and TabPFN-v2 (0.7269) is 0.0232 in favour of SAP-RPT-1, despite TabPFN-v2 leading in mean ROC-AUC. In Stage II, CatBoost has a lower reported ECE-10 than TabPFN-v2 at each displayed training size: 0.140 versus 0.180 at N = 25, and 0.040 versus 0.060 at full size. These contrasts show that discrimination, precision-recall quality, and calibration are complementary rather than interchangeable properties. They also caution against presenting the ROC-AUC result as evidence that the selected TFM is uniformly better on every aspect of predictive quality.

This distinction is especially relevant in operational settings. A batch-scoring workflow with a relaxed response-time requirement may place greater weight on discrimination and accept the additional inference cost. A high-throughput online service may prefer a GBDT when the latency advantage outweighs a modest reduction in mean ROC-AUC. Where probability quality drives downstream decisions, calibration should be evaluated or corrected independently of ranking performance. In practice, the appropriate choice is therefore a dataset- and use-case-specific trade-off rather than a universal model ranking.

6.3 Practical implications for model selection

For a task with a very small labelled training set, TabPFN-v2 is a strong candidate to include in the initial model comparison when ROC-AUC is a primary objective. Its mean

advantage is largest at N = 25, and the repeated paired results indicate that this lead is not simply the consequence of one favourable random subsample. This does not remove the need for task-specific validation: Phoneme and Sick exhibit early GBDT crossovers, showing that some datasets can behave differently even in the smallest evaluated regime. A short pilot comparison on the target dataset is therefore more defensible than assuming that a TFM will always win when labels are scarce.

When a larger labelled sample is available, the decision becomes more balanced. The mean ROC-AUC gap is smaller at N = 500, 750, 1,000, and full training size, and several GBDTs approach the TFM in the aggregate. At that point, the benefit of a small AUC difference should be weighed against the cost of inference and the volume of predictions to be served. Figure 3 can support this decision by showing which model-size combinations are non-dominated on the aggregate ROC-AUC and inference-time dimensions. It should be used as a comparison aid, not as a deployment recommendation independent of workload or service-level requirements.

Where calibrated probabilities are essential, the raw ECE-10 results suggest that the strongest-discrimination model should not be assumed to be the best probability estimator. A separate calibration assessment, and where appropriate a post-hoc calibration procedure, should be considered. Likewise, for class-imbalanced tasks, PR-AUC should be examined alongside ROC-AUC rather than inferred from it. These recommendations follow directly from the observed differences between the Stage-I ROC-AUC and PR-AUC rankings and the Stage-II ECE-10 results. They provide a practical interpretation of the benchmark without introducing a fixed rule for choosing a model from N alone.

The paired design holds test observations and sampled training rows constant across models within each condition. Five seeds measure sensitivity to the training subsample, while paired tests, multiplicity correction, effect sizes, and bootstrap intervals quantify differences within the benchmark. Because datasets and test partitions recur across training sizes, these results should not be read as independent replications across a population of tabular datasets.

Future work can extend this scaling design to additional TFMs, tasks, hardware profiles, and post-hoc calibration methods. The main design principles are to vary labelled sample size systematically, match data across competing models, repeat the sampling procedure, and report predictive outcomes together with resource requirements.

For deployment, model selection should begin with the available label volume and workload constraints, followed by matched comparisons at the sample sizes that matter operationally. ROC-AUC, PR-AUC, calibration, memory, and inference time should be considered together, and dataset-level outcomes should be reviewed before adopting an aggregate recommendation.

7. LIMITATIONS

The results are bounded by the evaluated datasets, tasks, models, and implementation choices. Stage II contains eight OpenML-CC18 binary-classification datasets, while Stage I compares two TFMs on 15 tasks. Multiclass classification, regression, distribution shift, and hybrid ensembles are outside the scope; only TabPFN-v2 is carried into the detailed scaling grid. The findings should therefore not be generalized to every TFM or structured-prediction task.

The train-test split is fixed for each dataset, and the five seeds vary only the training subsamples; variability from alternative held-out partitions is not measured. Conventional baselines undergo model-specific hyperparameter tuning using Optuna, and broader or differently configured searches could change their relative performance. The zero-shot TFM versus tuned-baseline comparison is practically relevant but does not use identical optimization budgets.

The dataset-selection rules reduce repeated representation of any one application domain but do not create a statistically representative sample of all tabular problems. Full training sizes also differ across datasets, so the aggregate full-data result summarizes dataset-specific maxima rather than one common N. Only two TFMs are compared in Stage I and only the selected model enters the detailed scaling grid; conclusions apply to these candidates and conditions, not to all foundation-model architectures.

Calibration is assessed on raw probabilities without post-hoc correction, and the datasets do not provide protected-class labels for formal demographic-fairness analysis. Memory and inference measurements depend on the recorded implementations and hardware. Finally, selecting TabPFN-v2 from its measured comparison with SAP-RPT-1 does not establish that it represents all current or future TFMs.

8. CONCLUSION

This two-stage benchmark examines the value of tabular foundation models when labelled training data are scarce and as task-specific supervision increases. Stage I provides an empirical basis for selecting TabPFN-v2 for detailed analysis. In Stage II, it achieves the highest mean ROC-AUC at all nine training sizes across eight datasets and five seeds. Its advantage over the strongest conventional baseline is largest at N = 25 (0.0346 AUC) and narrows overall to 0.0087 at full training size. Dataset-level crossover points nevertheless vary, precluding a universal threshold.

The wider evaluation shows why predictive performance must be interpreted alongside other outcomes. SAP-RPT-1 has higher mean PR-AUC than TabPFN-v2 in Stage I, CatBoost has the lowest reported ECE-10 in Stage II, and TabPFN-v2 incurs higher and increasing inference cost as context grows. TFMs are therefore particularly promising when labels are scarce and discrimination is important, while conventional learners may be preferable when latency, memory, or calibration requirements dominate.

Repeated paired evaluations, multiplicity-adjusted tests, effect sizes, and bootstrap intervals support the reported comparisons within the benchmark. The contribution is an empirical account of how predictive gains and deployment costs change as labelled tabular data accumulate, not a universal ranking of foundation and conventional models.

DATA AND CODE AVAILABILITY

The benchmark code, experiment configurations, detailed reports, results, and supporting figures are available at https://github.com/Siri1702/TabFM-Benchmark. The repository also provides more granular dataset-level and seed-level analyses and plots covering predictive performance, calibration, statistical comparisons, memory consumption, and inference time. These materials allow readers to inspect the underlying results and visual evidence beyond the aggregate summaries presented in this paper.

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